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Project 3 Aerodynamics

Abstract

This project expands on the source panel method developed previously by adding a point vortex and enforcing the Kutta condition to create a lift prediction for a NACA 4412 airfoil. The airfoil surface is discretized into source panels, and an additional unknown circulation is introduced through a vortex placed along the camber line. The strength of this vortex is determined by enforcing the Kutta condition at the trailing edge, allowing the model to produce physically realistic lifting flow.

Flow field visualizations of streamlines and velocity magnitude contours were generated for angles of attack at $\alpha = 0^\circ$, 10° , and 20° . Pressure coefficient distributions were solved and analyzed in relation to the velocity fields that were observed. The inclusion of circulation produced asymmetric flow behavior which was consistent with lifting airfoils, including increased velocity over the upper surface and a shifted stagnation point.

The circulation, Γ , and lift coefficient that corresponds, C_l , were calculated for each case and compared with thin airfoil theory. Results were then compared to Dr. Luca Virtuani's MATLAB XFOIL predictions, which were themselves derived from Prof. Mark Drela, for varying Reynolds numbers to analyze the influence of viscous effects. Compared to the previous source-only method, adding the vortex and Kutta condition significantly improved the accuracy of lift prediction and the behavior at the trailing-edge, though discrepancies do remain due to the assumption of inviscid flow.

Technical Approach

The source panel method was used to model inviscid, incompressible, irrotational flow over a NACA 4412 airfoil. This project builds on the source-only panel method by adding one point vortex and enforcing the Kutta condition. This change allows for the solver to predict lift, which the source-only method could not do accurately because it did not include circulation.

This process begins by defining the number of points on each side of the airfoil, the NACA airfoil number, the angle of attack, and the freestream velocity. For this project, the NACA 4412 airfoil was used with $V_\infty = 1$. The airfoil coordinates are generated using NACA_Coords_Tapered, and the airfoil surface is closed by adding the first point again at the end of the coordinate arrays. The panel centers are then calculated using.

$$x_i = \frac{X_i + X_{i+1}}{2}, \quad y_i = \frac{Y_i + Y_{i+1}}{2}$$

The function findUnitNormals is then used to calculate the panel spacing, tangential unit vectors, and normal unit vectors. These vectors are used throughout the code to project velocities into the normal and tangential directions.

The freestream velocity is broken into its x- and y- components using

$$U_\infty = V_\infty \cos \alpha, \quad V_\infty^{(y)} = V_\infty \sin \alpha$$

The no-penetration boundary condition is then applied at every panel center. This requires the normal component of velocity at the airfoil surface to be zero:

$$\vec{V} \cdot \hat{n}_i = 0$$

For the source panel method, this gives the linear system

$$\sum_{j=1}^N I_{ij} \lambda_j = - \vec{V}_\infty \cdot \hat{n}_i$$

where I_{ij} is the normal influence coefficient of source panel j on the control point i , and λ_j is the unknown source strength on panel j . The right hand side is defined in code as

$$RHS = - (U_\infty n_{x,i} + V_\infty^{(y)} n_{y,i})$$

Each panel center i and each source panel j are then iteratively calculated. For every pair of panels, two influence coefficients are calculated: I_{ij} , the normal velocity influence, and J_{ij} , the

tangential velocity influence. These are stored in matrix form and represent the geometric influence of each panel on every other panel.

The normal and tangential velocity influence are coded in an obfuscated function, but one method to find the influence of each source panel on each point, is to have code first solve for these values

$$\begin{aligned}
 A_{ij} &= - [(x_i - X_j) \cos \phi_j + (y_i - Y_j) \sin \phi_j] \\
 B_{ij} &= (x_i - X_j)^2 + (y_i - Y_j)^2 \\
 C_{ij} &= \sin(\phi_i - \phi_j) \\
 D_{ij} &= (y_i - Y_j) \cos \phi_i - (x_i - X_j) \sin \phi_j \\
 E_{ij} &= (x_i - X_j) \sin \phi_j - (y_i - Y_j) \cos \phi_i
 \end{aligned}$$

After solving, we can solve for the the normal influence integral, with the equation:

$$I_{ij} = \frac{C_{ij}}{2} \ln\left(\frac{S_j^2 + 2A_{ij}S_j + B_{ij}}{B_{ij}}\right) + \frac{D_{ij} - A_{ij}C_{ij}}{E_{ij}} \left[\tan^{-1}\left(\frac{S_j + A_{ij}}{E_{ij}}\right) - \tan^{-1}\left(\frac{A_{ij}}{E_{ij}}\right)\right]$$

The diagonal of the matrix is represented by π so that each panel can account for its own influence on itself, without causing a mathematical issue:

$$I_{ii} = \pi$$

The tangential velocity on the airfoil surface first requires solving for the integral of the j panel similar from before, resulting in the equation:

$$J_{ij} = \frac{D_{ij} - A_{ij}C_{ij}}{2E_{ij}} \ln\left(\frac{S_j^2 + 2A_{ij}S_j + B_{ij}}{B_{ij}}\right) - C_{ij} \left[\tan^{-1}\left(\frac{S_j + A_{ij}}{E_{ij}}\right) - \tan^{-1}\left(\frac{A_{ij}}{E_{ij}}\right)\right]$$

with the diagonal term set to

$$J_{ii} = 0$$

To include circulation, the one point vortex of unknown strength Γ is added, where the vortex is placed on the camber line at

$$x_\Gamma/c = \frac{1}{3}$$

For the NACA 4412 airfoil, the camber parameters are

$$m = 0.04, \quad p = 0.4$$

and the vortex y-location is calculated from the camber line equation:

$$y_c = \begin{cases} \frac{m}{p^2}(2px_\Gamma - x_\Gamma^2), & 0 \leq x_\Gamma \leq p \\ \frac{m}{(1-p)^2}[(1-2p) + 2px_\Gamma - x_\Gamma^2], & p < x_\Gamma \leq 1 \end{cases}$$

The velocity induced by a unit point vortex at each panel center is calculated using

$$u_\Gamma = -\frac{\Gamma(y_i - y_\Gamma)}{2\pi r_i^2}, \quad v_\Gamma = \frac{\Gamma(x_i - x_\Gamma)}{2\pi r_i^2}$$

where

$$r_i^2 = (x_i - x_\Gamma)^2 + (y_i - y_\Gamma)^2$$

These vortex velocity components are projected onto the normal and tangential unit vectors:

$$V_{\Gamma,i}^{(n)} = u_\Gamma n_{x,i} + v_\Gamma n_{y,i}, \quad V_{\Gamma,i}^{(t)} = u_\Gamma t_{x,i} + v_\Gamma t_{y,i}$$

The normal vortex influence is added as an extra column in the matrix because Γ is an additional unknown. This changes the no-penetration equation to

$$\sum_{j=1}^N I_{ij} \lambda_j + \Gamma V_{\Gamma,i}^{(n)} = -\vec{V}_\infty \cdot \hat{n}_i$$

Adding circulation represents the rotation of the flow, which allows for the solution to satisfy the physical requirement for lift. Since adding Γ creates one extra unknown, the one extra equation using the Kutta condition must be implemented. The Kutta condition requires the flow to leave the trailing edge smoothly, so the tangential velocity jump at the trailing edge should be removed. This is enforced using the upper and lower trailing edge panels:

$$V_{t,TE_1} + V_{t,TE_2} = 0$$

The source contribution to this condition is formed using the tangential influence matrix

$$\sum_{j=1}^N (J_{TE_1,j} + J_{TE_2,j}) \lambda_j$$

The vortex contribution is

$$\Gamma(V_{\Gamma,TE_1}^{(t)} + V_{\Gamma,TE_2}^{(t)})$$

and the freestream tangential contribution is

$$U_{\infty}(t_{x,TE_1} + t_{x,TE_2}) + V_{\infty}^{(y)}(t_{y,TE_1} + t_{y,TE_2})$$

This produces the final Kutta row:

$$\sum_{j=1}^N (J_{TE_1,j} + J_{TE_2,j})\lambda_j + \Gamma(V_{\Gamma,TE_1}^{(t)} + V_{\Gamma,TE_2}^{(t)}) = - [U_{\infty}(t_{x,TE_1} + t_{x,TE_2}) + V_{\infty}^{(y)}(t_{y,TE_1} + t_{y,TE_2})]$$

The final system is put in a matrix system:

$$\begin{bmatrix} I_{ij} & V_{\Gamma}^{(n)} \\ J_{TE_1,j} + J_{TE_2,j} & V_{\Gamma,TE_1}^{(t)} + V_{\Gamma,TE_2}^{(t)} \end{bmatrix} \begin{bmatrix} \lambda_j \\ \Gamma \end{bmatrix} = \begin{bmatrix} -\vec{V}_{\infty} \cdot \hat{n}_i \\ -V_{\infty,TE}^{(t)} \end{bmatrix}$$

As previously mentioned, the addition of the vortex term and Kutta condition changes the original $N \times N$ system into a $(N + 1) \times (N + 1)$ system, where the final unknown is the circulation required to produce lift. This matrix is solved, where the first N entries of the solution vector are the source strengths λ_j , and the final value is the vortex circulation Γ .

After solving λ_j and Γ , the code calculates the surface velocity. The normal and tangential velocity components are formed by adding the freestream contribution, source panel contribution, and vortex contribution:

$$V_n = \vec{V}_{\infty} \cdot \hat{n} + I\lambda + \Gamma V_{\Gamma}^{(n)}, \quad V_t = \vec{V}_{\infty} \cdot \hat{t} + J\lambda + \Gamma V_{\Gamma}^{(t)}$$

The total surface velocity magnitude is then

$$V = \sqrt{V_n^2 + V_t^2}$$

and the pressure coefficient is calculated

$$C_p = 1 - \left(\frac{V}{V_{\infty}}\right)^2$$

This directly relates velocity to pressure through Bernoulli's equation, allowing regions of acceleration and deceleration along the airfoil to be interpreted in terms of pressure changes. Upper and lower surfaces are separated and the C_p plotted in code. The lift coefficient is then calculated from the vortex strength using

$$C_l = \frac{L'}{\frac{1}{2}\rho V_\infty^2 c}$$

Where substituting the Kutta-Joukowski's equation for lift per unit span:

$$L' = \rho V_\infty \Gamma$$

This relationship shows that lift is directly proportional to circulation, which links the solution of the panel method to the physical lifting flows. A non-dimensional chord is used, where $c = 1$, leaving the equation:

$$C_l = -\frac{2\Gamma}{V_\infty}$$

The negative sign comes from the convention used in the code, where a negative Γ corresponds to the circulation direction and that it produces positive lift for this airfoil.

Thin air foil theory is compared to the calculated lift coefficient, where the camber line slope is used to calculate the thin airfoil coefficients A_0 and A_1 , and the lift coefficient is calculated from a fourier series expansion, where thin airfoil theory defines the circulation of a cambered line as:

$$\Gamma = V_\infty c \left(\pi A_0 + \frac{\pi}{2} A_1 \right)$$

where

$$A_0 = \alpha - \frac{1}{\pi} \int_0^\pi \frac{dz}{dx} d\theta$$

$$A_1 = \frac{2}{\pi} \int_0^\pi \frac{dz}{dx} \cos \theta d\theta$$

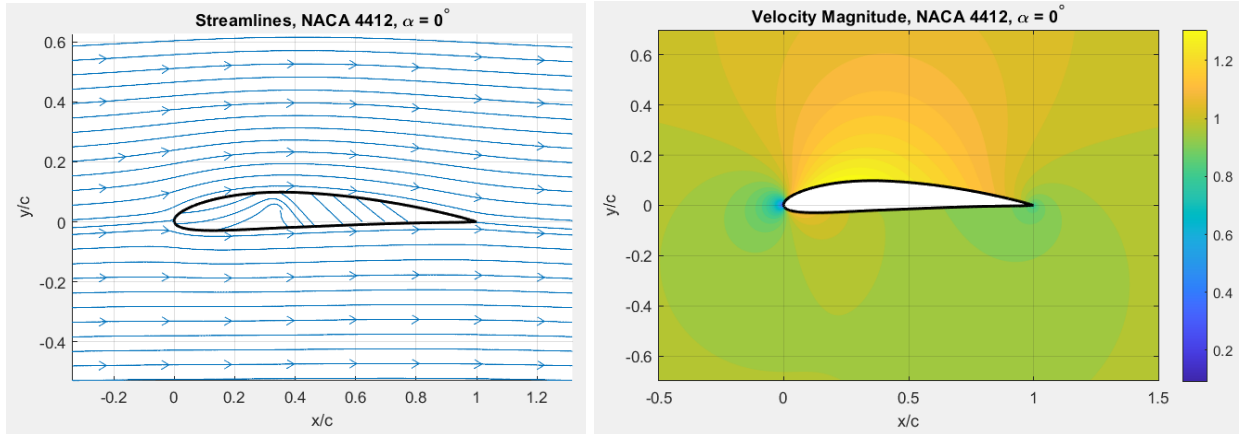
Plugging Γ into our equation for C_l ,

$$C_l = \pi(2A_0 + A_1)$$

This solution provides a baseline for comparison, as it assumes a thin cambered plate and neglects thickness, which allows differences with the panel method to highlight the influence of the airfoils geometry.

Task 1 Results

For all simulations, 1,000 panels were used.



Figures 1 & 2. Streamlines and velocity magnitude contours for the NACA 4412 airfoil at angles of attack $\alpha = 0^\circ$

At $\alpha = 0^\circ$, the flow is mostly smooth around the airfoil but is not perfectly symmetric because the NACA 4412 airfoil is cambered. The streamlines bend around the upper surface more noticeably than the lower surface which shows that the airfoil still produces circulation even at zero angle of attack. In the velocity magnitude plot, higher velocity appears over the upper surface, while lower velocity appears near the leading edge and trailing edge. The low velocity near the leading edge represents a stagnation region, where the flow slows as it meets the airfoil surface.

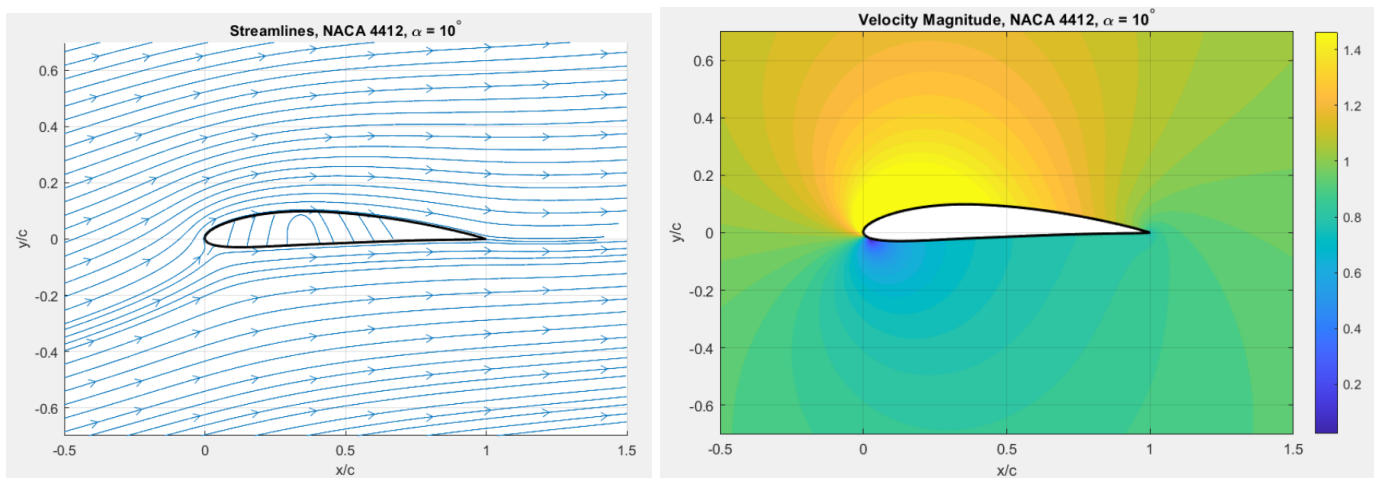


Figure 3 & 4. Streamlines and velocity magnitude contours for the NACA 4412 airfoil at angles of attack $\alpha = 10^\circ$

At $\alpha = 10^\circ$, the flow becomes more strongly asymmetric. The streamlines are deflected upwards before reaching the airfoil and bend more over the upper surface. This shows the stronger turning effect caused by the increased angle of attack and the added circulation from the vortex. The velocity magnitude contour shows a larger high-velocity region near the lower leading edge. This happens because the flow accelerates over the upper surface where the streamline spacing becomes tighter, while the flow slows near the stagnation point on the lower/front portion of the airfoil.

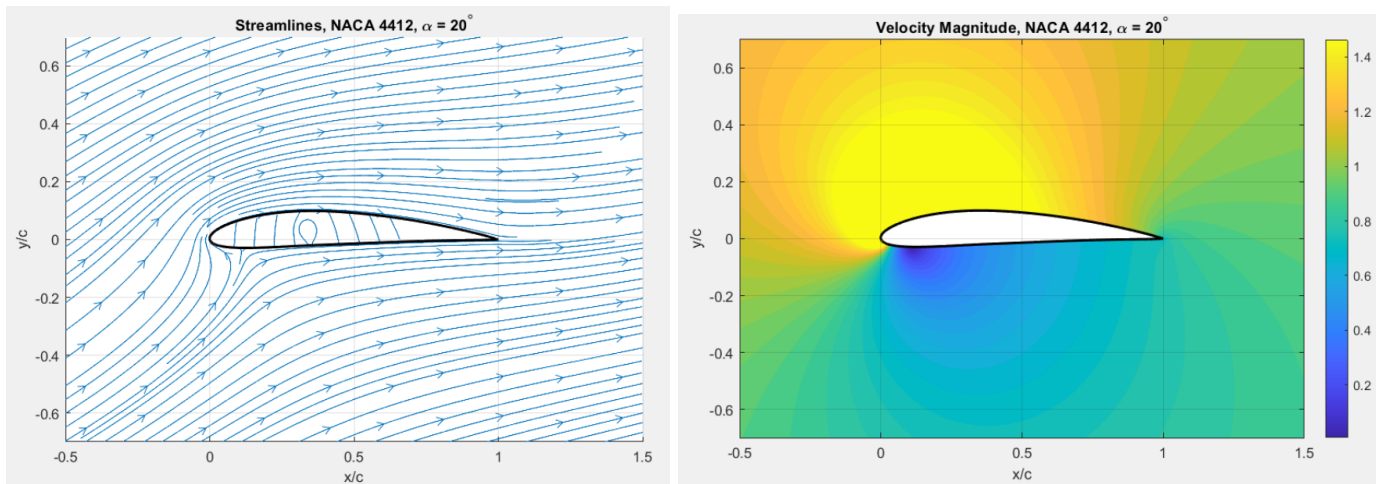
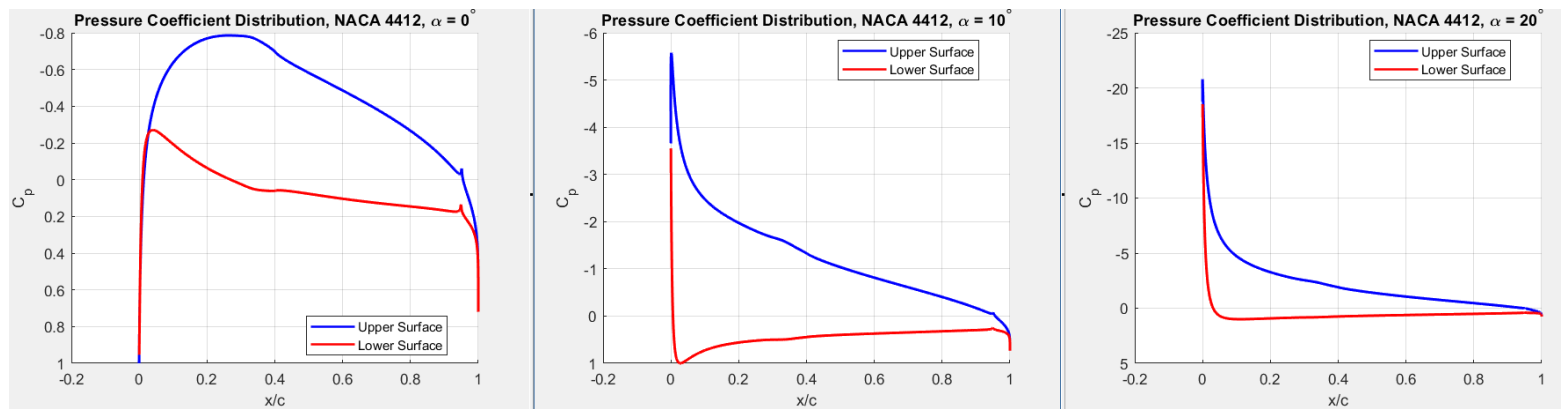


Figure 5 & 6. Streamlines and velocity magnitude contours for the NACA 4412 airfoil at angles of attack $\alpha = 20^\circ$

At $\alpha = 20^\circ$, the effect becomes even stronger. The streamlines show greater curvature around the airfoil, especially near the leading edge and over the upper surface. The velocity magnitude plot shows the highest velocities above the airfoil and lower velocities below it. The low-velocity region near the lower leading edge becomes more noticeable because the incoming flow meets the airfoil at a steeper angle, moving the stagnation region farther downward. The higher velocity over the upper surface is caused by the flow accelerating around the cambered surface and the increased circulation required to satisfy the Kutta condition.

The increase in the angle of attack also increases the asymmetry of the flow field. The upper surface experiences higher velocity because the flow is turned more strongly and the streamlines are closer together. The lower velocity regions occur near the stagnation points, especially around the leading edge and lower surface, where the flow slows before moving around the airfoil. Compared to Project 2, the addition of the vortex and Kutta condition produces a smoother and more physical trailing-edge flow, because the tangential velocity is forced to leave the trailing edge consistently rather than producing the unrealistic behavior that can be seen in the source-only method.



Figures 7, 8, & 9. The pressure coefficient distribution for the NACA 4412 airfoil at $\alpha = 0^\circ$, 10° , & 20° , respectively.

At $\alpha = 0^\circ$, the pressure distribution is not symmetric because the NACA 4412 airfoil has camber. The upper surface has a lower C_p than the lower surface for most of the chord, which means the upper surface has a higher velocity. This matches the velocity magnitude contour, where the region above the airfoil showed higher flow speed than the lower surface. The lower surface has a higher C_p , meaning lower velocity and greater pressure compared to the upper surface.

At $\alpha = 10^\circ$, the difference between the upper and lower surfaces becomes much larger. The upper surface shows a strong negative C_p near the leading edge, which represents a sort of peak in suction. This correlates with the velocity magnitude contour, where the highest velocities occurred over the upper/front portion of the airfoil. The lower surface stays at a higher C_p , showing that the flow is slower behind the airfoil. This pressure difference between the upper and lower surfaces is what produces increased lift.

At $\alpha = 20^\circ$, the pressure coefficient difference becomes even more extreme. The upper surface has a very large suction peak near the leading edge, while the lower remains much closer to positive or slightly negative C_p . This matches the velocity magnitude contour, where the highest velocity region was above the airfoil and the lower velocity region was below the airfoil. Since C_p is defined as:

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2$$

a higher velocity produces a lower, more negative C_p , while a lower local velocity produces a higher C_p . This is why the upper surface, where the flow accelerates, has the lowest pressure coefficient values.

Increasing the angle of attack causes a stronger acceleration over the upper surface and a larger pressure difference between the upper and lower surfaces. The C_p plots support the velocity magnitude contours because the areas with high velocity correspond to low C_p , while the areas with low velocity correspond to high C_p . This trend shows why lift increases as angle of attack increases with the panel method.

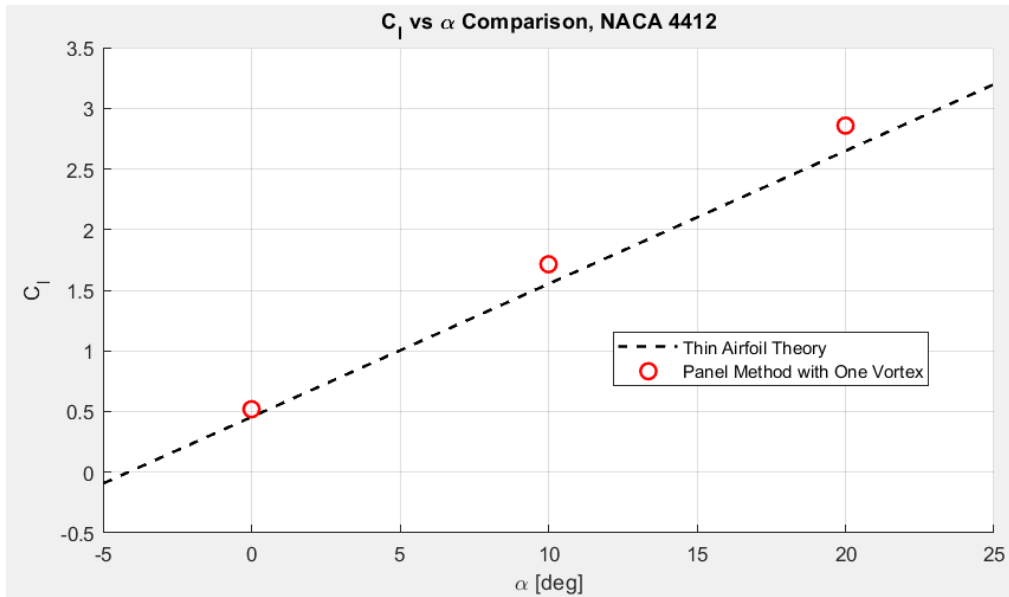


Figure 10. Lift coefficient, C_l , versus angle of attack, α , for the NACA 4412 airfoil, comparing the source panel method with one vortex to thin airfoil theory.

For each angle of attack, the source panel method was solved with one added vortex. The vortex strength, Γ , was then used to calculate the lift coefficient using

$$C_l = -\frac{2\Gamma}{V_\infty}$$

Since $V_\infty = 1$, this becomes $C_l = -2\Gamma$.

Table 1. Circulation and lift coefficient for the NACA 4412 airfoil at different angles of attack.

Angle of Attack	Circulation, Γ	Lift Coefficient, C_l
0°	-0.259705	0.519410
10°	-0.857143	1.714286
20°	-1.428537	2.857074

Figure 10 shows the comparison between the panel method with one vortex and thin airfoil theory. The panel method follows the same general linear trend as thin airfoil theory, where

C_l increases as angle of attack increases. This is expected because increasing α increases the circulation required to satisfy the Kutta condition, which then increases lift.

The panel method predicts slightly higher lift than thin airfoil theory at all three angles of attack. At $\alpha = 0^\circ$, the NACA 4412 still produces positive lift because it is cambered, so the zero-lift angle is negative rather than zero. As α increases to 10° and 20° , both methods continue to show increasing lift, but the panel method points stay above the thin airfoil theory line.

The circulation values are negative because of the sign convention used in the vortex's creation. In the coordinate system, a negative value of Γ corresponds to the direction of circulation needed to produce positive lift on the airfoil. That is why the lift coefficient is calculated using -2 , making the resulting C_l positive for all three angles of attack.

The main difference is that the thin airfoil theory assumes a thin cambered plate and does not fully account for airfoil thickness. The panel method uses the full NACA 4412 geometry, including both camber and thickness. Based on the plotted comparison, adding thickness increases the predicted lift for this case, since the panel method gives larger C_l values than thin airfoil theory at the same angles of attack. However, this is still inviscid, so it does not include separation or stall, meaning the 20° result is likely higher than what would occur in real viscous flow.

Task 2 Results

To further determine the accuracy of the source panel method with an added vortex, results were compared to those obtained using XFOIL. XFOIL is an advanced panel method that uses additional physical effects such as boundary layer development, transition, and viscous interactions. Unlike the inviscid panel solver used in this project, XFOIL accounts for the influence of viscosity on the flow, allowing for more realistic predictions of pressure distribution, lift, and drag.

In this section, calculations found from panel method predictions are compared with XFOIL results for both inviscid and viscous cases at different Reynolds numbers. This is done to show limitations of the source/vortex panel method and the role of viscous effects.

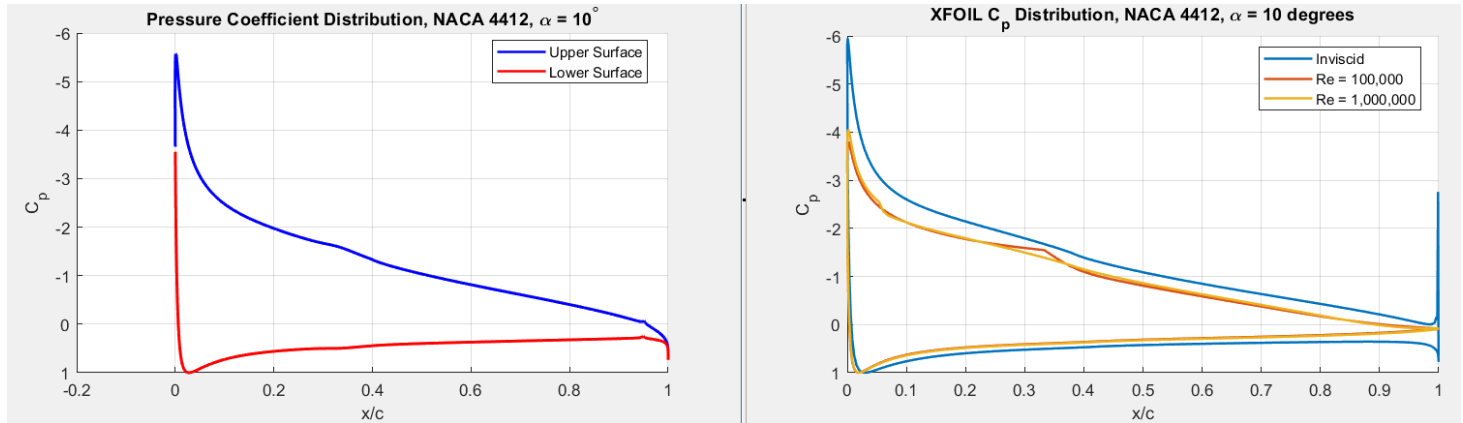


Figure 11 & 12. Compared pressure coefficient distribution for the NACA 4412 airfoil at $\alpha = 10^\circ$. Figure 11 shows the result from the panel solver with one vortex, while Figure 12 shows the XFOIL results for the inviscid case, $Re_c = 100,000$, and $Re_c = 1,000,000$.

The panel solver and XFOIL results show the same general trend. In all cases, the upper surface has a strong negative C_p near the leading edge, which represents the suction peak and high velocity over the upper surface. The lower surface has a much higher C_p , meaning the flow is slower and pressure is greater below the airfoil. This agrees with the panel solver's results, where the largest velocities occur over the upper surface.

The main difference is that the panel solver predicts a slightly different pressure recovery along the chord. The panel method has a strong leading-edge suction peak and then a gradual recovery toward the trailing edge. The XFOIL inviscid case shows the closest overall behavior to the panel solver because both methods are based primarily on inviscid potential flows. However, the XFOIL inviscid result has a sharper suction peak and slightly different behavior near the trailing edge.

The viscous XFOIL cases, $Re_c = 100,000$, and $Re_c = 1,000,000$, show lower suction over much of the upper surface compared to the inviscid case. This happens because the viscous effects and boundary layer development reduce the acceleration near the surface. Between the two viscous cases, the $Re_c = 1,000,000$ curve is slightly closer to the inviscid result, while the $Re_c = 100,000$ case shows a larger viscous effect.

The inviscid XFOIL case is closest to the panel solver because both neglect viscosity. The viscous XFOIL cases differ more because they include boundary layer effects, transition, and separation behavior which the source panel method with one vortex does not model.

As Re increases from 100,000 to 1,000,000, the C_p distribution moves slightly closer to the inviscid XFOIL result. The higher Reynolds number case shows a stronger suction region on the upper surface and a smoother recovery towards the trailing edge. This happens because increasing Re reduces the effect of viscosity, making the flow behave more like the inviscid case.

In the XFOIL results, there is a noticeable change in the slope of the upper surface C_p curve around the mid chord region near $x/c \approx 0.35$ to 0.40 . This is likely a point where the boundary layer transitions, where the flow changes from laminar to turbulent. The $Re_c = 100,000$ case shows the transition effect more noticeably, while the $Re_c = 1,000,000$ case appears closer to the inviscid curve and transitions earlier.

Compared to the expected flat plate transition, this makes sense. For a smooth flat plate, transition is expected at around $Re_x \approx 5 \cdot 10^5$. Since $Re_x = Re_c (x/c)$, a chord Reynolds number of 100,000 wouldn't likely reach the required Reynolds number for transition over the chord. For $Re_c = 1,000,000$, transition would be expended near

$$\frac{x}{c} = \frac{Re_x}{Re_c} = \frac{5 \cdot 10^5}{1 \cdot 10^6} = 0.5$$

which is close to the transition behavior seen in the middle of the airfoil. However, because the airfoil has pressure gradients and suction peak, the transition location does not exactly have to match the flat plate. The adverse pressure gradient over the upper surface can cause transition to occur earlier than the ideal flat plate estimate.

Table 2. Comparison of lift coefficient and drag coefficient for panel method and XFOIL cases at $\alpha = 10^\circ$.

Case	Lift Coefficient, C_l	Drag Coefficient, C_d
Panel method w/ vortex	1.71429	0
XFOIL inviscid	1.83376	0
XFOIL $Re_c = 100,000$	1.38319	0.02620
XFOIL $Re_c = 1,000,000$	1.40855	0.01670

The inviscid XFOIL case predicts the highest lift coefficient, while the panel method gives a slightly lower but similar value. Both inviscid methods predict zero drag because viscosity is not included, and in potential flow there is no skin-friction drag or pressure drag from separation.

When viscosity is included in the XFOIL cases, the lift coefficient decreases. At $Re_c = 100,000$, C_l drops to 1.38319, and at $Re_c = 1,000,000$, it increases slightly to 1.40855 but still is below the inviscid value. This shows that viscosity reduces lift by slowing the near-wall flow, thickening the boundary layer, and changing the effective pressure around the airfoil.

Viscosity also creates drag. The inviscid cases have $C_d = 0$, while the viscous XFOIL cases produce nonzero drag values. The lower Reynolds number case has higher drag because viscous effects are stronger, while at the higher Reynolds number, drag decreases since the boundary layer is thinner and viscous losses are reduced.

The viscosity decreases lift and increases drag compared to the inviscid solutions. As Reynolds number increases, the results move closer to the inviscid case, with slightly higher lift and lower drag.

Summary & Conclusions

This project builds on the source panel method that was previously developed by introducing a point vortex and enforcing the Kutta condition to model lifting flow over a NACA 4412 airfoil. The source-only panel method was able to represent the geometry of the airfoil and produce reasonable pressure distributions, but it could not accurately predict lift due to the absence of circulation. By adding a vortex, the method was extended to include circulation, allowing for realistic lift predictions and improved trailing edge behavior. As shown throughout the results, the addition of the vortex produced asymmetric flow fields, higher velocities over the upper surface, and pressure differences consistent with lifting airfoils.

From these methods, it became apparent that circulation strongly influences aerodynamic performance. The relationship between velocity, pressure, and lift was directly observed through the connection between velocity magnitude contours and pressure coefficient distributions. The increase in angle of attack led to stronger suction and larger pressure differences, which resulted in increased lift. Additionally, comparing the panel method to thin airfoil theory showed that including airfoil thickness leads to slightly higher predicted lift, which shows the importance of geometry in analysis.

The comparison with XFOIL further showed limitations of the inviscid panel method. While the panel method and inviscid XFOIL produced similar trends, including viscosity in XFOIL introduced boundary layer effects, reduced lift, and generated drag. These differences emphasized that real aerodynamic flows are influenced by viscous effects such as transition and separation, which are not captured by potential flow. However, even with these limitations, the panel method was a good method for showing the fundamental behavior of how flow interacts with an airfoil.

Overall, these panel methods are insightful, especially for understanding the physics of aerodynamic flows. They provide an efficient way to analyze how geometry and angle of attack affect pressure distribution and lift. The main lesson learned is that even simplified models can capture the main principles of a problem when correct assumptions are applied, such as enforcing the Kutta condition. At the same time, it is important to know when to use other methods for viscous effects.

This extends beyond aircraft design as understanding how flow accelerates around surfaces, how pressure distributions form, and how circulation generates forces is applicable to many engineering systems like turbines or vehicle aerodynamics analysis. Interpreting flow behavior using both simplified and advanced methods is important for designing efficient and effective aerodynamic systems in a wide range of applications.

Appendix

- Anderson, J. D., *Introduction to Flight*, 6th Edition, McGraw-Hill Education.
- Virtuani, L., “[Xfoil for MATLAB](#),” MATLAB Central File Exchange.
- Drela, M., “[XFOIL: An Analysis and Design System for Low Reynolds Number Airfoils](#),” MIT.
- Zigunov, A., “[Source Panel Method Plus Vortex](#),” Syracuse University.